Temasek Polytechnic
School of Informatics & IT
Diploma in Applied Artificial Intelligence
Robotic Process Automation (CAI2C01)
AY2026/2027 April Semester

Project Specification (100%)

Introduction

The objective of this project is to understand and apply the knowledge of Robotic Process Automation to automate end-to-end processes of an organisation/company. You are to identify, design, and create a robot for a real-life use case for a business scenario.

You may consider, but are not limited to, the following scenarios:

1. Sales Order Entry Robot
2. Stock Market Robot (Buy & Sell stocks)
3. Automatic Email Responder Robot
4. Invoice Processing Robot
5. Data Migration Robot
6. HealthCare Administration (Patient Data Entry and Scheduling)
7. Insurance Claims Processing
8. Supply Chain and Inventory Management (Low Inventory Alerts and Order Placements)
9. Job Application Processing Robot

Project Components

The project consists of **FIVE** deliverables listed the table below:

Deliverable	Percentage	Submission	Due Date
Project Proposal	20%	TPLMS	28th May, Thursday, 1700 hrs
Project Deliverables Application Part A	20%	TPLMS	9th July, Thursday, 1700 hrs
Project Deliverables Application Part B	30%	TPLMS	3rd Aug, Monday, 1700 hrs
Project Deliverables Report	10%	TPLMS	6th Aug, Thursday, 1700 hrs
Project Demonstration	20%	In classroom	Week 18 (check with your tutor)

Note:

a) This is an **individual** project.
b) Discuss the project proposal business scenario with your tutor and get approval.
c) Project weightage is 100%.

Project Scope

Identify, design, and create a robot for any real-life business domain e.g. human resources, online retail or customer service. You may choose any business domain. There are no restrictions on the type of robot, as long as it is feasible within the timeframe.

Your robot solution should interact with at least ONE application.

Examples of applications (non-exhaustive list):

1. MS Outlook
2. MS Excel
3. PDF
4. Web browser
5. Image file (using AI-powered OCR to extract text)
6. MS Word/Word Processing
7. Python code
8. Offline Systems
9. Database
10. Connecting to Other Systems

Project Proposal (20%)

Submit a project proposal report, in either word/pdf format **within 8 pages** excluding the cover page, declaration of originality and references by **28th May, Thursday 1700hrs** via TPLMS under Assessments folder. You can name the document **2420202A_ProjectProposal.docx or 2420202A_ProjectProposal.pdf.** Multiple submissions are allowed. Only the latest will be graded.

Please use the cover page provided in TPLMS to include your name and admission number. Your proposal should include the following sections:

Robot Name	Give a name for your robot
Business Domain	Provide description of the business domain E.g. ITSM/Banking/Education/Marketing/Sales/
Target Department/Users	Explain who will be using this automation. It can be department(s) in an organisation or specific user group(s)
Project Scope	Description of your robot with purpose and functions
Justification	Need for automation Value of automation How does it improve productivity?
Robot Category	Attended/Unattended
Current Scenario	List and explain the steps involved in the current scenario (Before automation).
Automation Scenario	List and explain the robot execution steps.
Process Flow Chart	Flow chart to illustrate the automation solution. The flow chart should have the application used at each step.
References	Web URL/Books/Research papers/Citations of literature

Table 1 Proposal Requirements

Project Deliverables Application Part A (20%)

Requirements for Project Deliverables Application Part A:

- Robot Application Program Code
  - One most significant and working application (e.g. reading from PDF files) that is fully tested
  - At this stage, the remaining proposed applications need not work.
  - Identify and provide the significant activities needed to support the remaining applications as specified in your proposal. The remaining applications do not have to work. The activities for the remaining applications do not have to be configured using the properties panel at this point in time.
- Any input datasets or output resources (e.g. PDF files, Image files) for your project that supports the one most significant application.

You need to zip the project folder with any datasets/resources and submit it to TPLMS, named in the following format: *MatricNo_FullName_ProjectDeliverablesPartA.zip* e.g. **2420202A_BruceLee_ProjectDeliverablesPartA.zip, by 9th July, Thursday, 1700 hrs**.

Please present your Project Deliverables Application Part A during Week 13 to your tutor. Tutor may provide formative feedback during this session. Please ensure you have a backup copy of your files in case there is a problem with the online submission. Please do not use other file compression methods (e.g. RAR). Multiple submissions are allowed. Only the latest will be graded.

Project Deliverables Application Part B (30%)

Requirements for Project Deliverables Application Part B:

- Complete robot application that is fully tested (project folder with source code)
- Any input datasets or output resources (e.g. PDF files, Image files) for your project

You need to zip the project folder with any datasets/resources and submit it to TPLMS, named in the following format: *MatricNo_FullName_ProjectDeliverablesPartB.zip* e.g. **2420202A_BruceLee_ProjectDeliverablesPartB.zip, by 3rd Aug, Monday, 1700 hrs**.

Please ensure you have a backup copy of your files in case there is a problem with the online submission. Please do not use other file compression methods like RAR. Multiple submissions are allowed. Only the latest will be graded.

Project Deliverables Report (10%)

- Report (10%)
  - Presentation slides – **maximum of 10 slides** excluding title and references slides – use RPA project report slide template from the TPLMS project folder.
  - Presentation slides should include screenshots and explanation of **all** packages and activities you have used.
  - Presentation slides should also include a YouTube URL link to the explanation video that does **NOT exceed 10 mins**. Please turn on your camera when you voice over your demonstration. Make sure you set the video to unlisted.

You need to submit the Project Deliverables Report to TPLMS, named in the following format: *MatricNo_FullName_ProjectDeliverablesReport.ppt* e.g. **2420202A_BruceLee_ProjectDeliverablesReport.ppt, by 6th Aug, Thursday 1700hrs**.

Please use the provided template available on TPLMS. Please ensure you have a backup copy of your files in case there is a problem with the online submission. Multiple submissions are allowed. Only the latest will be graded.

Project Demonstration (20%)

Demonstrate and present your robot with your report on **Week 18 (check with your tutor)**. Please keep your demo to be NO more than 10 mins. During the demo, the tutor may ask you questions or make modifications to your work to test your understanding.

Using Generative AI for your proposal

You can use Generative AI tools for your proposal by using appropriate prompts. The following are some examples:

- "Generate 3 ideas for an RPA project to automate invoice processing in an aluminum can manufacturing factory."
- "What are some steps to automate in invoice processing in an aluminum can manufacturing factory?"
- "Provide 3 RPA project ideas for streamlining our customer onboarding process in a hotel stay."
- "How can RPA help us reduce manual data entry in our customer onboarding process in a hotel stay?"

Student Guidelines in using Generative AI Tools

- You are responsible for your own work.
- The AI is unpredictable.
- The results may not be always correct.
- Do not copy and paste the exact results. You are required to significantly modify from it.
- Cite where you have used Generative AI tools.

You can also refer to page 32 of the ChatGPT Playbook Student Edition provided in TPLMS → Assessment section for effective use of Generative AI Tools. Also refer to page 7 of what NOT to do with ChatGPT.

Frequently asked questions

Q: I have submitted my project proposal already. Can I still change the project domain and proposal?

A: No. It would be good not to change the domain after the proposal has been submitted. The project was released early. So, think and decide the domain carefully.

Q: I have proposed to use 2 applications - PDF and Excel in the robot, in the project proposal.

Now, can I change applications to Excel and Outlook or include more applications?

A: Please discuss with your tutor. The tutor can approve or disapprove the change depending on the time needed for you to change, taking into consideration the deadline of project.

Q: The instructions indicate that at least 1 application is to be used in the robot. Will I be able to pass the module if I use ONLY ONE application?

A: Using only one application meets the minimum requirement, but marks depend on other rubric criteria. Please refer to the rubrics for the various OTHER factors on how marks will be awarded.

Q: Will I get more marks if I use more applications?

A: We want you to explore more applications interacting with your bot to show your understanding of the different functionalities. Please refer to the rubrics on how marks will be awarded.

Q: Rubrics mentions interesting functions. What are interesting functions?

A: Basic functions may include but are not limited to opening a file and reading or writing data to it. Interesting functions may include but are not limited to extracting specific data using regex, sending email with body of email extracted from notepad/customized body of email with professional formatting etc.

Q: Can I use any website for registration for users for my problem statement?

A: Some websites may not allow bots to work on their websites. In the past, we have had students who tried working with some websites but resulted in their IP address being banned after few tries. Please also note certain websites continuously change the name/id of the textbox or label as a security feature making it hard to read or write to the textbox without errors. It may be better to create your own webpage in such cases.

Q: Can I create a new website on my own for my problem statement?

A: Yes, but no additional marks will be provided for the website creation.

Some points to take note of while using websites:

- Sliding button may be there, instead of clicking buttons for registrations
- Advertisements may pop up
- "Accepting cookies" may pop-up
- Robots may not be allowed to register
- May get banned after few tries by robot
- The coupon codes may change – if you are scraping them
- The name/id of the textboxes on webpages and labels may change every few minutes
- The price from the E-commerce websites cannot be taken from the main page of data scraping. You must click separately and find the price on that page.

Q: Why is there a need to demonstrate and present when I have recorded a video and uploaded to YouTube?

A: The video recording will **not** be used as a substitute for demonstration. The recording is just to give us an idea on the scope of your project. We can only provide marks based on what we observe during the face-to-face demonstration.

Q: I got the idea for my project from YouTube. Can I use it?

A: Please check with your tutor first. Upon his approval, please cite your references in your submission. You are also required to significantly improve uponeeyour cited work/ source of reference. See Academic-related Offences.

Penalty for Late Submission

- Late within 1 day : 10% deduction from absolute mark given for the assignment
- Late > 1 day and up to 2 days : 20% deduction from absolute mark
- Late > 2 days : 0 marks awarded

Note that "day" includes non-working days (Saturday, Sunday, and public holidays).
General MC/LOA is NOT considered as valid reason for extension of assignment submission.

Academic-related Offences

Academic-related Offences	Recommended Punitive actions
○ **Cheating on Semestral Examination** (including possession of unauthorised materials as verified by the Examiner as relevant to the examination paper) ○ **Cheating on Coursework** such as projects, term tests and assessments ○ **Plagiarism** Please refer to the Plagiarism Policy	Punitive actions could include the following: o Warning letter o Failing the assessment item/subject o Suspension from course of study for a minimum period of one semester o Dismissal from the Polytechnic

Here are some resources that you might find useful:

- Introduction to creating flowcharts (7 mins 22 sec) https://www.youtube.com/watch?v=SWRDqTx8d4k
- How to make a flowchart: https://www.technokids.com/blog/programming/how-to-make-a-flowchart-for- programming-easy-to-understand/
- How to create stunning flowcharts in Microsoft Word (4 mins 27 sec) https://www.youtube.com/watch?v=iiS7aAFI2Cs
- How to upload video to YouTube: https://support.google.com/youtube/answer/57407

Annex C: Marking Criteria

Criteria	In Context	A >=80%	B 70-79%	C 60-69%	D 50-59%	F <50%
Project Proposal (20%)	Refer to Table 1 Proposal Requirements (20 marks)	- All required items are provided with explanations that provide clarity and meaning to the proposal. - Project Scope and Justification is supported by strong and valid reason, evidence, and visual aids - Current/Automation scenario is explained in detail as step-by-step procedure. - Process flowchart has detailed steps with application mentioned where needed.	- All required items are provided with some much explanation - Purpose of Robot is supported by valid reasons, evidence, or visual aids - Current/ Automation scenario is explained in detail - Process flowchart has sufficient steps with some details.	- Some required items are provided without much explanation - Purpose of the robot is clear but largely unsupported by reason, evidence, or visual aids - Current/Automation scenario is very brief - Process flowchart has brief steps.	- Only some of the required items are provided - Purpose of the robot is vague and largely unsupported by reason, evidence, or visual aids - Current/Automation scenario is incomplete - Process flowchart has very brief steps.	- Non-submission of all required items - Clear evidence of plagiarism detected (disciplinable offence) - (or) - Many of the required items are missing - Purpose, style of writing and presentation impedes reader's understanding - Current/Automation scenario is incomplete - Process flowchart present but incomplete.

Project Deliverables Application Part A (20%)	Error Handling (5 marks)	The RPA effectively anticipates and handles errors with comprehensive error detection, logging, and recovery mechanisms in place.	The RPA adequately handles errors with basic error detection and recovery mechanisms, ensuring minimal disruption to automated processes.	The RPA inconsistently handles errors, lacking robust error detection and recovery mechanisms, resulting in occasional disruptions to processes.	The RPA struggles to handle errors, often causing significant disruptions to processes with limited error detection and recovery capabilities.	The RPA fails to handle errors effectively, leading to frequent disruptions and failures in automated processes due to inadequate error management.
	Executable (5 marks)	Executes without error	Executes without error but may have trivial issues (eg. Warnings)	Executes with minor errors	Executes with some error	Robot does not run
	Problem-solving (10 marks)	The RPA effectively addresses the identified problem, offering innovative and efficient solutions that meet or exceed project objectives and requirements.	The RPA adequately solves the identified problem, providing solutions that meet most project objectives and requirements with moderate innovation.	The RPA partially addresses the identified problem but falls short in providing comprehensive solutions or innovative approaches to meet project objectives.	The RPA's problem-solving efforts are ineffective, failing to adequately address the identified problem and missing key project objectives.	The RPA's problem-solving approach is severely deficient, completely missing the mark in addressing the identified problem and project objectives.

Project Deliverables Application Part B (30%)	Execution (10 marks)	Robot interacts with or uses more than 4 different applications Provide interesting functions for each of the applications, beyond opening a file, and read & write all data to the file. Process is efficiently organised making it easy to follow. Sequence and variable names have a very clear relationship to their purpose in the process. Works independently with minimal guidance; effectively uses resources. Able to apply significant and new knowledge, beyond what was taught in class.	Robot interacts with or uses 4 different applications Provide interesting functions for most applications, beyond opening a file, and read & write all data to the file. Process is well-organised making it easy to follow. Sequence and variable names have a partial relationship to their purpose in the process. Works independently with occasional guidance; uses resources well.	Robot interacts with or uses 3 different applications Provide interesting functions for some applications, beyond opening a file, and read & write all data to the file. Process is organised making it easy to follow. Sequence and variable names have limited relationship to their purpose in the process. Often needs guidance; limited use of resources.	Robot interacts with or uses 1 or 2 different applications Provide basic functions for each application. Process is not organised making it difficult to follow. Sequence and variable names have no relationship to their purpose in the process. Relies heavily on guidance; minimal or ineffective use of resources.	Robot interacts with or uses 0 applications Provide some basic functions. Process is messy without proper names for sequences and variables.
	Error handling (5 marks)	Executes without error Provided error handling activities using nested try-catch and validation checking wherever needed.	Executes without error but may have trivial issues (eg. Warnings) Provided error handling activities and validation checks in 80% of the places needed.	Executes with one error Provided error handling activities and validation checks in 60% of the places needed.	Executes with two error Provided error handling activities and validation checks in 50% of the places needed.	Executes with some error Provided very few error handling activities or validation checks.

	Robot is as per proposal (5 marks)	Robot is created as per proposal and able to automate all steps proposed	Robot is created as per proposal and able to automate more than 80% of steps proposed	Robot is created as per proposal and able to automate more than 60% of steps proposed	Robot is created as per proposal and able to automate more than 50% of steps proposed	Robot is created as per proposal and able to automate less than 50% of steps proposed
	Use of Orchestrator (5 marks)	Provide good and relevant use of unattended or scheduled bot with Orchestrator. Able to apply new knowledge, beyond what was taught in class and explores the use of triggers such as API Triggers.	Provide relevant unattended or scheduled bot with Orchestrator that implements advanced triggers.	Provide unattended or scheduled bot with Orchestrator that implements advanced triggers.	Provide unattended or scheduled bot with Orchestrator that implements simple time triggers.	Provide incomplete configuration for unattended or scheduled bot with Orchestrator.
	Annotations/ Comments (5 marks)	Well commented/ annotated for all the activities used	Well commented/ annotated for most of the activities used.	Meaningful comments/ annotations provided for only some activities used.	Meaningful comments/ annotations provided for only important activities used.	Meaningful comments or annotations for the activities are mostly missing.

Project Deliverables Report (10%)	Process (1 mark)	Provide very clear and thorough details of the process.	Provide mostly clear and thorough details of the process.	Provide a limited list of steps of the process.	Provide a few steps of the process.	Does not provide the process.
	Packages (2 marks)	Provide screenshots of all the packages used.	Provide screenshots of most of the packages used.	Provide screenshots of a limited number of packages used.	Provide screenshots of a few of the packages used.	Does not provide screenshots of the packages used.
	Explanation of flow of process with activities used (7 marks)	Provide very clear and thorough explanation of the flow of the process with activities used, and packages used.	Provide mostly clear and thorough explanation of the flow of the process with activities used, and packages used.	Provide a some explanation of the flow of the process with activities used, and packages used.	Provide very limited explanation on the process with activities used, and packages used.	Provide no explanation on the process, packages, and activities.

Project Demonstration (20%)	Able to explain the application (10 marks)	Demonstration captures all the work done with appropriate pace, sequencing, and visual aids.	Demonstration captures most parts of the work done. -	- Demonstration captures some parts of the work done. -	- Demonstration captures very few parts of the work done. -	- Absent for project presentation and demonstration without a valid reason. (or) - Demonstration fails to capture the work done. -
	Able to answer questions or making changes to the bot (5 marks)	Able to answer all the questions.	Able to answer most of the questions.	Able to answer some of the questions.	Able to answer very few questions.	Not able to answer questions.
	Keep to allocated time (5 marks)	Length of demo matches the allocated time of 10 minutes.	Length of demo time is exceeded by 2 minutes.	Length of demo time is exceeded by 3 minutes.	Length of demo time is exceeded by 4 minutes.	Length of demo time is shorter than / exceeded by 5 minutes.

***** END OF PROJECT SPECIFICATION *****